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Investigating the effect of urban form on land surface temperature at block and grid scales based on XGBoost-SHAP

Hongfei Li [a,b], Jun Yang [a,c,d,*], Jiaxing Xin [d], Wenbo Yu [d], Jiayi Ren [d], Huisheng Yu [b,e], Xiangming Xiao [f], Jianhong (Cecilia) Xia [g]

[a] *JangHo Architecture College, Northeastern University, Shenyang, 110169, China*
[b] *Liaoning Key Laboratory of Urban and Architectural Digital Technology, JangHo Architecture College, Northeastern University, Shenyang, 110169, China*
[c] *Human Settlements Research Center, Liaoning Normal University, Dalian, 116029, China*
[d] *School of Humanities and Law, Northeastern University, Shenyang, 110169, China*
[e] *School of Management Engineering, Qingdao University of Technology, Qingdao, CO, 266520, China*
[f] *School of Biological Sciences, University of Oklahoma, Norman, OK, 73019, USA*
[g] *School of Earth and Planetary Sciences (EPS), Curtin University, Perth, 65630, Australia*

A R T I C L E I N F O

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A B S T R A C T

The urban thermal environment is becoming increasingly severe. In this study, we integrated eXtreme Gradient Boosting with the SHapley Additive exPlanations method to investigate the effects of various urban factor indexes (UFIs) on land surface temperature (LST) at both block and grid scales. Additionally, we examined the differences in LST and its driving factors across local climate zones (LCZs) at the grid scale. The results show that LST is higher in central areas than in peripheral ones during summer and autumn, but this pattern is reversed in spring and winter. LST varies significantly across LCZs, with the normalized difference built-up index, normalized difference vegetation index (NDVI), and Shannon's diversity index (SHDI) identified as the main contributors. The sky view factor inhibits LST at the block scale but promotes it at the grid scale. The impacts of UFIs follow the seasonal trend: summer > spring > autumn > winter. LST responses to UFIs exhibit similar trends across scales, showing specific warming or cooling thresholds—for example, a cooling effect when SHDI exceeds 0.65, and a warming effect when building density exceeds 20 % (summer and autumn) or 40 % (spring and winter). Significant cooling occurs only when NDVI exceeds 0.4; however, NDVI generally remains low in all seasons except summer. High-contribution UFIs typically exhibit the strongest interaction effects with artificial factor indicators.

1. Introduction

The Sixth Assessment Report of the Intergovernmental Panel on Climate Change states that global average temperatures are now 1.1 °C above pre-industrial levels. This rise in global temperature has significantly impacted natural ecosystems and health (Trisos et al., 2020), with prolonged warming and climate change leading to biodiversity loss and increasing the frequency and intensity of extreme weather events (Ham et al., 2023; Kafy et al., 2024). A recent report in *The Lancet* stated that heat-related deaths among individuals over 65 years of age have risen by 85 % globally since 1990 (Romanello et al., 2023). One major consequence of global warming is the rise in the surface temperature, which is particularly evident in urban areas. The land surface temperature (LST) is the most direct indicator of the urban thermal environment (Peng et al., 2016), and urban sprawl increases the proportion of hard-paved surfaces and concrete buildings, limiting the dissipation of absorbed solar heat and intensifying the local LST (Zhou et al., 2022). This phenomenon adds complexity to the urban thermal environment. Currently, more than 50 % of the global population resides in urban areas, and this figure is projected to reach 68 % by 2050 (Han et al., 2024) implying that more people will be exposed to the severe threats posed by the increase in LST in the future. Understanding the drivers of LST changes is crucial if thermal environment threats are to be addressed.

Investigating the relationship between LST and its driving factors is

* Corresponding author. JangHo Architecture College, Northeastern University, Shenyang, 110169, China.
E-mail addresses: 2301565@stu.neu.edu.cn (H. Li), yangjun8@mail.neu.edu.cn (J. Yang), 2310012@stu.neu.edu.cn (J. Xin), 2110013@stu.neu.edu.cn (W. Yu), 2210011@stu.neu.edu.cn (J. Ren), xiangming.xiao@ou.edu (X. Xiao), c.xia@curtin.edu.au (J.(C. Xia).

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essential for developing effective strategies that can address the challenges of the urban thermal environment (Chaudhuri and Mishra, 2016). Most previous studies investigating the LST have focused on its relationship with land use/land cover (Diren-Üstün et al., 2024; Yuan et al., 2023). Urban spatial patterns are typically characterized by impervious surfaces, and the elemental and morphological variations within cities are often overlooked (Eldesoky et al., 2021). The local climate zone (LCZ) classification scheme effectively addresses these differences in both urban-rural and intra-urban settings (He et al., 2023). Stewart and Oke (2012) introduced this categorization in their study of urban thermal environments, linking urban form with thermal properties. Different types of LCZs can be identified by integrating the surface features in a three-dimensional (3D) space, providing a feasible standard framework for identifying the driving factors of LST by taking differences in urban internal spatial form as the starting point (Huang et al., 2023).

Different quantitative characterizations of urban form yield distinct results regarding the driving effects on LST. However, an excessive number of urban factor indexes (UFIs) will inevitably lead to redundant description, and regression results may be distorted due to indicator collinearity (Ding et al., 2025). Therefore, the quantification of urban space should involve the selection of appropriate indicators from a simple perspective that can directly guide the adjustment of urban artificial and natural forms (Peng et al., 2018). The urban form in the study area is dominated by large parks, farmlands, buildings, and pocket parks, which exhibit both horizontal expansion and vertical growth. Therefore, we focused on selecting element indicators that can describe these spatial forms. Urban artificial elements increase the absorption of thermal radiation and affect the transfer of heat (Lu et al., 2021). The normalized difference built-up index (NDBI) is a clear indicator that can distinguish between artificial and natural underlying surfaces in urban areas; it can increase LST by absorbing thermal radiation (Cetin et al., 2024). Mean Building Height (MBH), Building Density (BD), Sky View Factor (SVF), and Floor Area Ratio (FAR) are typically used to quantify the horizontal and vertical configuration of urban buildings (Guo et al., 2016; Yang et al., 2018), with the Proportion of Construction Land (PCL) and Proportion of Bare Soil (PSB) further reflecting the horizontal distribution of construction and non-construction land (Guo et al., 2020). Natural elements also significantly influence the urban thermal environment by absorbing solar thermal radiation and altering energy circulation (Bai et al., 2024). The normalized difference vegetation index (NDVI) can describe the overall distribution of vegetation (Rodríguez-Gómez et al., 2025), while Shannon's diversity index (SHDI) enables the further refined differentiation of vegetation types, effectively distinguishing the differences in vegetation diversity among various natural elements (Liu et al., 2024). Patch density (PD), the contagion index (CONTAG), and the landscape shape index (LSI), are also highly correlated with LST, respectively describing the spatial distribution of small-area natural vegetation, the connectivity between green space units, and the shape complexity of green space (Ding et al., 2023). The proportion of woodland and grass (PWG) describes the horizontal spatial distribution of vegetation, while mean tree height (MTH) supplements the vertical distribution. In contrast, the population density (POP) reflects the horizontal aggregation degree associated with human activities (Ma et al., 2024). Although previous studies have contributed to the development of large datasets describing spatial distribution, there is no consensus regarding the perception of urban building forms shaped by artificial factors in the urban thermal environment (Lemoine-RodrÃ et al., 2022). Therefore, further verification and analysis are still required to formulate regionally differentiated spatial control strategies (Su et al., 2021).

Early studies mostly explored the correlation between LST and UFIs or used multiple linear regression to analyze the quantitative relationship between the two (Sekertekin and Zadbagher, 2021). However, this relationship is not simple nor linear (Li and Hu, 2022). Advances in research resulted in the introduction of the random forest (RF) method, which can be used to investigate the non-linear relationship between the urban space and the LST from a global perspective (Mirsanjari et al., 2021), while methods such as multiscale geographically weighted regression (MGWR) and geographically weighted regression have been applied to explore the spatial heterogeneity between LST and the urban spatial form from a local perspective (Kashki et al., 2021; Yin et al., 2022). With the development of technology, the eXtreme Gradient Boosting (XGBoost) model has begun to replace traditional regression models in urban thermal environment research due to its high operational efficiency and more effective handling of missing values (Bushenkova et al., 2024). However, pure machine learning models suffer from the "black box" problem. To better interpret the regression results, the SHapley Additive exPlanations (SHAP) method has been introduced and combined with the XGBoost model to more accurately explain the relationships between explanatory variables and their impacts on the thermal environment (Cheng et al., 2025), with the combination of the two addressing the interaction effects that previous models were unable to explore (Liu et al., 2025), enabling the better investigation of the complex effects of UFIs on LST and the interrelationships among different UFIs (Luo et al., 2023).

The research scale directly affects the applicability of the results, with strategies for different research conclusions varying across different scales, from global to intra-regional, inter-urban, and intra-urban (Liang et al., 2022). Grid scales that match the data accuracy are often used in research into the intra-urban thermal environment due to the format of the data provided (raster images); however, this method randomly fragments urban spaces that are integral in nature (Yelixiati et al., 2024). Additionally, scale sensitivity occurs in the relationship between the LST and UFIs (Yao et al., 2020), further prompting comparison between different grid scales (Liu et al., 2021). However, a disconnection between the research results and basic urban units (blocks) remains. The irregular block scale, with its objective boundaries, can effectively align with urban management units, making up for the shortcomings of using grids in terms of spatial segmentation and practical application (Zhu et al., 2023). Nevertheless, it is still necessary to integrate research on the two scales to reduce the uncertainties inherent in single-scale results, especially at the interaction level, for which sufficient samples from the grid scale are required for in-depth exploration (Han et al., 2023).

To clarify the differences between block and grid scales, the area within the Shenyang 4th Ring Road Expressway in China was selected as a study area, with the aims of: (1) Constructing an urban LCZ model to quantify and illustrate the spatiotemporal heterogeneity of LST across different LCZs; (2) using the LCZ framework to rigorously assess the seasonal contributions of UFIs to LST across various LCZ types on a 100 m $\times$ 100 m grid scale; and (3) exploring the effects of different UFIs on LST and analyzing the response of LST to UFIs on both block and 100 m $\times$ 100 m grid scales. By quantitatively describing the urban spatial form and transforming complex urban systems into specific and controllable element indicators, the research results can guide urban planning practices, with the specific goal of alleviating the urban thermal environment and improving the quality of human settlements by controlling UFIs. The heterogeneity between the LCZ and LST provides a spatial reference for formulating urban thermal mitigation policies, which is crucial for managing and controlling the future urban spatial form in undeveloped areas and transforming the spatial form in urban areas.

2. Study area and data sources

2.1. Study area

Shenyang is located in northeastern China and is the capital of Liaoning Province (between 122°25′9″ and 123°48′2″ E, and 41°11′51″to 43°2′13″ N). It has a temperate continental climate. The study area for this research was the central urban region within Shenyang's Fourth Ring Road, encompassing nine municipal districts: Shenhe, Heping, Dandong, Huanggu, Tiexi, Sujiatun, Hunnan, Shenbei New Area, and

Yuhong, with a total population of 5,750,534 as of 2020 (https://tjj.shenyang.gov.cn/). This area serves as the core of Shenyang's political, economic, and cultural activities. Urban construction in this region is concentrated and well-developed, featuring diverse landscapes ranging from urban to suburban, with varied land cover types and complex thermal environmental characteristics. Due to the presence of planned but undeveloped block construction sites within the Fourth Ring Road, this region is ideal for studying the impacts of various UFIs on LST. The location of the study area in Shenyang is shown in Fig. 1.

2.2. Data sources

This study used remotely sensed imagery from Landsat 8 TIRS/OLI, along with OpenStreetMap (OSM) data for roads, land use, buildings, a digital elevation model (DEM), mean tree height (MTH), population data (POP), and administrative boundaries. Landsat 8 TIRS imagery was used to derive LST data for all four seasons, while Landsat 8 OLI imagery was used to calculate the NDVI and NDBI. Land use data were employed to calculate landscape pattern indexes (PD, SHDI, LSI, PWG, PSB, PCL, and CONTAG). Building vector data were used to compute MBH, BD, and FAR, while DEM data were used to calculate the SVF. OSM road data were used to extract the Fourth Ring Road to define the scope of the study area. POP reflected the degree of human agglomeration within each unit, and tree height data were used to obtain MTH for the study units. Table 1 lists the data sources and their descriptions.

3. Methodology

The methods adopted in this study involved data collection and processing, indicator calculation and screening, and contribution and response analyses (Fig. 2). First, block units were delineated based on road network data, and a geographic information system (GIS) method was applied to generate LCZs at two scales: a 100m × 100m grid and irregular blocks. Next, LST data for the four seasons were derived from remote sensing images, and relevant artificial and natural factor indicators were selected and calculated. All raster data were standardized to a resolution of 100m × 100m, while vector data were calculated separately according to the cell sizes at the grid and block scales. Then, the variations in the contributions of UFIs across different LCZs at the grid scale were analyzed. Finally, we compared the seasonal differences in LST across the LCZs. Subsequently, regression analysis was conducted using the XGBoost model, with UFIs as the independent variables and LST as the dependent variable. The regression results from the XGBoost model were interpreted using the SHapley Additive Explanations (SHAP) method, followed by a comparative analysis to evaluate the differences in the influence of LST and various UFIs at both the 100m × 100m grid and block scales.

Table 1
Data used and sources.

Data	Time	Resolution	Source link	Data source
Landsat8 TIRS Landsat8 OLI	2020 2020	100m 30 m	https://earthexplorer.usgs.gov/	U. S. Geological Survey (USGS)
Population data	2020	100 m	https://hub.worldpop.org/	Worldpop
DEM data	2020	30 m	https://www.gscloud.cn/	Geospatial Data Cloud
Tree height	2019	30 m	https://glad.umd.edu/	Global Land Analysis and Discovery
Land use data	2020	10 m	https://livingatlas.arcgis.com/	Esri|Sentinel-2 Land Cover Explorer
Road data	2020	vector	https://www.openstreetmap.org/	Open Street Map
Build data	2020	vector	https://data.tpdc.ac.cn/	National Tibetan Plateau Data Center
Administrative boundary	2020	vector	https://www.webmap.cn/	National Catalogue Service For Geographic Information

3.1. LST data retrieval

Remote sensing images for all four seasons of 2020, each captured during specific time phases, were acquired from the Landsat 8 satellite. Image frames with less than 5 % cloud cover were selected to minimize interferences from cloudiness. We employed composite images, the elements of which were averaged to create an average surface temperature for the season. The methodology entailed the initial removal of cloud-obscured regions and outlier regions. Subsequently, surface temperature values were averaged over multiple periods to compute the final LST product. The radiative transfer equation (RTE) shares a similar principle with the split-window-driven temperature-and-emissivity separation method: both reduce estimation errors by simplifying parameter inputs (Wang et al., 2024). The RTE exhibit higher accuracy

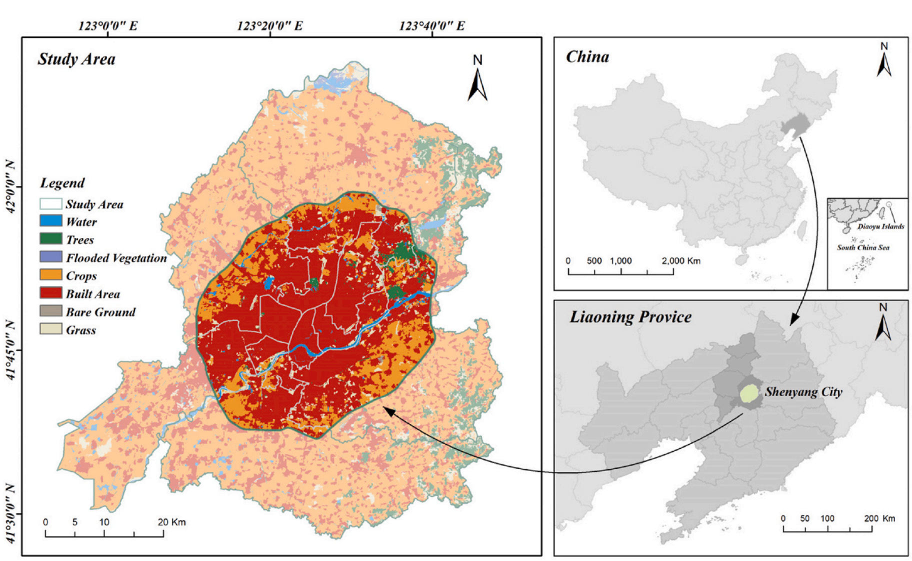

Fig. 1. Location of the study area in Shenyang.

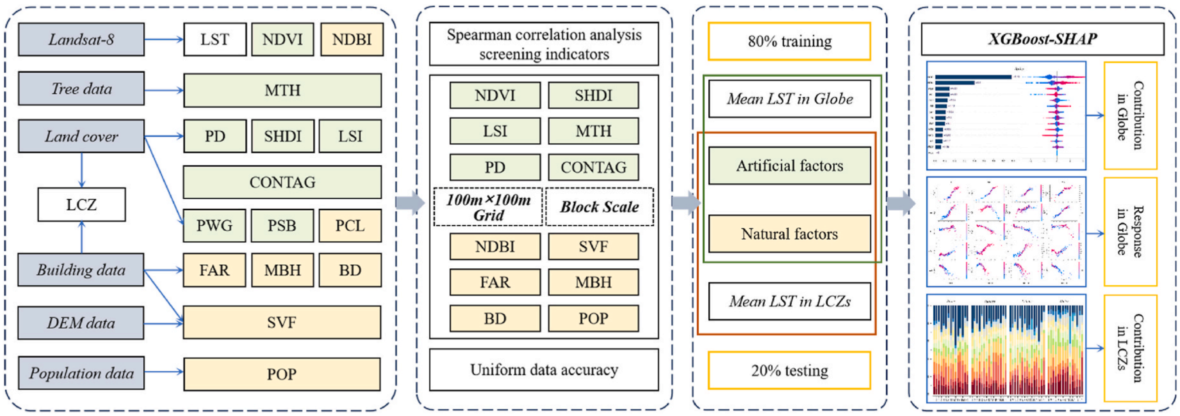

Fig. 2. Study methodology.

in retrieving LST from Landsat 8 data compared to the split-window algorithm and the single-channel method, with the RMSE lower than 1 K (Yu et al., 2014). The strictly calibrated Landsat 8 LST data has been verified to have an accuracy range of 0.5 K to 1 K at various terrestrial sites (Crawford et al., 2023). We applied the RTE method for LST inversion using the following formula:

$$L_\lambda = [\varepsilon \cdot B(T_S) + (1-\varepsilon)L_\downarrow] \cdot \tau + L_\uparrow \tag{1}$$

where L_λ is the satellite's radiant brightness, ε is the surface-specific emissivity, T_S is the true surface temperature (K), $B(T_S)$ represents the blackbody thermal radiant brightness, $L_\downarrow$ is the atmospheric upward radiant brightness, τ is the atmospheric transmittance, and $L_\uparrow$ is the atmospheric downward radiant brightness.

The blackbody thermal radiant brightness $B(T_S)$ in the thermal infrared band for an LST of T_S can be calculated using the following formula:

$$B(T_S) = \frac{[L_\lambda - L_\uparrow - \tau \cdot (1-\varepsilon)L_\downarrow]}{\tau \cdot \varepsilon} \tag{2}$$

T_S was derived using Planck's formula:

$$T_S = \frac{k_1}{\ln\left(\frac{k_2}{B(T_S)} + 1\right)} \tag{3}$$

where k_1 and k_2 are constants ($k_1 = 774.8853$ and $k_2 = 1321.0789$) in the Landsat 8 data.

The inversion results at 30m × 30 m resolution were converted from raster to points within the block units and calculated using zonal statistics. The LST for each block unit was determined by averaging the LST points within each respective block. The same methodology was applied to determine LST at the 100m × 100m grid scale. Landsat 8 TIRS captures the 100m LST product. To match the 30 m resolution of OLI data, the USGS applied atmospheric correction and downscaling resampling to the TIRS data to produce a 30m resolution LST product.

3.2. LCZ mapping

Commonly used methods for LCZ mapping include GIS, RS, and combined RS-GIS approaches (Liu et al., 2023; Quan and Bansal, 2021). Among these, the RS-based methods show greater accuracy in mapping land-cover types, while GIS-based methods are more effective for mapping built-type LCZs (Aslam and Rana, 2022). However, RS-based methods cannot handle irregular grids, so we chose the GIS method for LCZ mapping (Secerov et al., 2015). First, impervious surfaces were extracted as the primary calculation range of built-type LCZ, and land-cover-type LCZs were identified using other land use data. Second, based on the zoning reference index proposed by Stewart and Oke (2012) and considering the specific conditions in Shenyang, LCZ classifications on both the block and grid scales were determined in ArcGIS Pro 3.0 using land use and building data (Table 2), allowing definition of the built-type LCZs on both scales. Finally, a "raster to point" operation was performed on the land use data in ArcGIS Pro 3.0 and the "zonal statistics" tool used to calculate proportions within the grid and block units to assess the land-cover-type LCZs. The results of the LCZ mapping on both block and grid scales are shown in Fig. 3.

At present, most researchers choose 100m × 100m as the ideal scale for LCZ studies at the urban level (Quan and Bansal, 2021), as it ensures that buildings are not overly fragmented by dense grids while maintaining a high level of research accuracy. Before conducting LCZ classification on the block scale, a secondary subdivision of the suburbs was performed based on vegetation and landforms to compensate for any potential misclassifications caused by a sparse road network. However, one limitation of this method is that some smaller LCZ types could not be easily identified. To ensure consistency in the comparative analysis between the block and grid scales, LCZ types with insufficient sample sizes were thus excluded from the study. The final LCZ classifications included 13 distinct types, with LCZ2 (compact midrise) the most common built-type LCZ and LCZD (low plants) the most common land-cover-type LCZ.

3.3. eXtreme Gradient Boosting-SHapley Additive exPlanation (XGBoost-SHAP) method

LST is influenced not by a single UFI but by a complex interplay between multiple morphological factors (Li et al., 2021). Regression analysis was employed to examine the compound effects of these various morphological factors. In this study, the mean LST was designated as the dependent variable, while the UFIs were treated as independent variables. After all variables were fully normalized, the impact of each UFI on the LST was assessed using the XGBoost model. The dataset was randomly split, with 80 % allocated for training and 20 % reserved for testing. Model accuracy was evaluated using the coefficient of determination (R^2). Finally, the regression model was optimized using "GridSearchCV" (Chang and Lin, 2011).

The SHAP method was employed to interpret the results of the XGBoost regression. This analysis revealed the relative importance of each UFI to the LST. However, it did not indicate whether the influence was positive or negative. The SHAP method effectively addresses the "black box" issue that is inherent in the XGBoost model (Lundberg et al., 2020), explaining the output of the model by assigning importance values to the features. Both the magnitude and direction of UFI impacts were visualized by generating a summary plot using the SHAP package and inputting the data into Python.

Table 2
Classification basis of LCZ types.

LCZ types	PCL (%)	BD (%)	SVF	MBH (m)	MTH (m)	Land use types
LCZ1 (Compact high-rise)	40–60	40–60	0.2–0.4	>25	–	Impervious surface
LCZ2 (Compact midrise)	30–50	40–70	0.3–0.6	10–25	–	Impervious surface
LCZ3 (Compact low-rise)	20–50	40–70	0.2–0.6	3–10	–	Impervious surface
LCZ4 (Open high-rise)	30–40	20–40	0.5–0.7	>25	–	Impervious surface
LCZ5 (Open midrise)	30–50	20–40	0.5–0.8	10–25	–	Impervious surface
LCZ6 (Open low-rise)	20–50	20–40	0.6–0.9	3–10	–	Impervious surface
LCZ7 (Lightweight low-rise)	<20	60–90	0.2–0.5	2–4	–	Impervious surface
LCZ8 (Large low-rise)	40–50	30–50	>0.7	3–10	–	Impervious surface
LCZ9 (Sparsely built)	<20	10–20	>0.8	3–10	–	Impervious surface
LCZ10 (Heavy industry)	20–40	10–30	0.6–0.9	5–15	–	Impervious surface
LCZA (Dense trees)	<10	<10	<0.4	–	3–30	trees
LCZB (Scattered trees)	<10	<10	0.5–0.8	–	3–15	trees
LCZC (Bush, scrub)	<10	<10	0.7–0.9	–	<2	trees
LCZD (Low plants)	<10	<10	>0.9	–	<1	Grass、Crops
LCZE (Bare rock or paved)	>90	<10	>0.9	–	<0.25	Impervious surface
LCZF (Bare soil or sand)	<10	<10	>0.9	–	<0.25	Bare ground
LCZG (Water)	<10	<10	>0.9	–	–	Flooded vegetation、Water

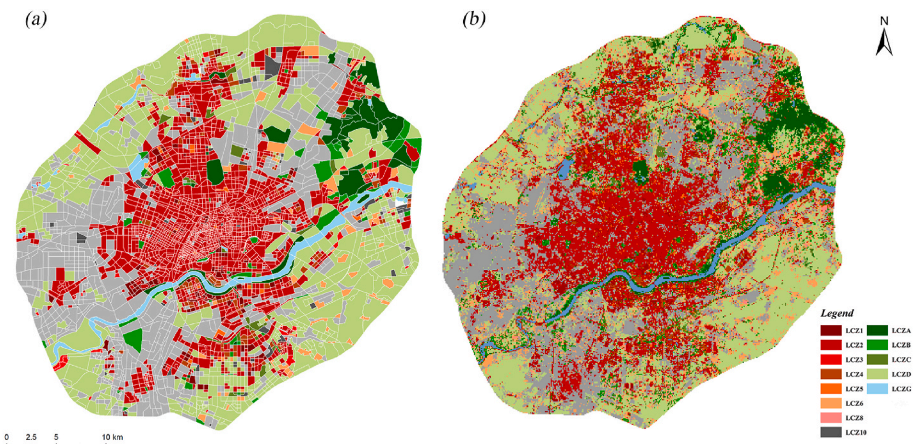

Fig. 3. LCZ map of the Shenyang 4th ring at the (a) block and (b) grid scales.

3.4. UFI visualization

Exploring the driving mechanisms that influence LST is crucial for optimizing the urban thermal environment. Of the several factors contribute to the spatial and temporal differentiation of LST,14 were selected and divided into two categories (artificial and natural factors) in this study, allowing quantification of the differences on both scales (Table 3). MBH, BD, SVF, FAR, PCL, POP, and NDBI were selected as artificial factor indicators to accurately measure the impacts of artificial UFIs on LST. We hypothesized that natural UFIs also play an important role in reducing the LST and selected NDVI, MTH, SHDI, PD, CONTAG, LSI, and PWG to explore the influence of natural factors. UFI values were derived on both the block and the 100 m × 100 m grid scales following a series of preprocessing steps, including multisource data correction, computation, and data cleaning.

In practice, data often suffers from redundancy, noise, and imbalance, rendering it necessary to preprocess the data before constructing any model. In addition, a high degree of correlation between independent variables leads to more serious data redundancy, affecting the fitting performance of a model. Spearman correlation analysis was used to eliminate redundant variables, with those that were significantly correlated with LST at the $p < 0.01$ level retained. In terms of the remaining indicators, if the Spearman correlation coefficient between any two indicators exceeded 0.8, one of the two was eliminated and only one retained for subsequent study. As shown in Fig. 4, this process resulted in the retention of 12 out of the original 15 indicators: MBH, BD, SVF, FAR, POP, NDBI, NDVI, MTH, SHDI, PD, CONTAG, and LSI.

4. Results

4.1. Spatiotemporal heterogeneity of LST in LCZs

Fig. 5 illustrates the seasonal distribution differences in the LST. Shenyang is located in a high-latitude region; thus, the cold season lasts longer than it does in other areas, with spring and winter exhibiting similar LST distribution characteristics, and cooler low-temperature zones in the city center as compared to the surrounding areas. This pattern is due to the colder conditions during spring and winter, reduced vegetation cover on suburban farmland (LCZD), and increased heat absorption by bare land. In summer, the built-up area (built-type LCZ) displays high LST values, while the surrounding unbuilt areas (land-cover-type LCZ) maintain relatively low LSTs. The largest area with high LST values is also observed in summer. In autumn, low LST values appear more dispersed, with localized clusters of high temperatures. By contrast, in winter, LST values are lower in built-up areas than unbuilt areas and tend to be more concentrated, while natural areas such as the farmland in the northwest and southeast regions exhibit higher LST values. Interestingly, the LST differs significantly upstream and downstream in the Hunhe River (LCZG), with the downstream segment demonstrating higher LST values than those observed upstream. The northeastern suburbs are dominated by mountainous forests (LCZA and LCZB) composed of tall natural coniferous trees, which help maintain a low LST throughout the year.

The box plots in Fig. 6 indicate notable differences in the LST across the various LCZs. Although outliers are observed in the LST values of

Table 3
Natural/Artificial factors indicators and description.

Category	Indicators	Formula	Unit	Description
Natural factors indicators	Patch density (PD)	$PD = \frac{N}{A}$	–	Density per unit area of landscape patches
	Landscape shape index (LSI)	$LSI = \frac{N}{A}$	–	The overall length of greenspace relative to the smallest possible edge length for a fully aggregated class within a unit area
	Contagion index (CONTAG)	$CONTAG = 1 + \frac{\sum_{i=1}^{n}\sum_{k=1}^{n}\left[P_i\left(\frac{g_{ik}}{\sum_{k=1}^{n} g_{ik}}\right)\right]\left[\ln P_i\left(\frac{g_{ik}}{\sum_{k=1}^{n} g_{ik}}\right)\right]}{2\ln(n)}$	–	The extent of aggregation or expansion of different patch types within a given landscape unit
	Mean trees height (MTH)	$MTH = \frac{\sum_{i=1}^{n} T_i}{n}$	m	Mean height of trees in unit area
	Shannon's diversity index (SHDI)	$SHDI = -\sum_{i=1}^{m}(P_i \ln P_i)$	–	The proportion of each patch type is multiplied by the total of all proportions within the unit area
	Normalized difference vegetation index (NDVI)	$NDVI = \frac{NIR - R}{NIR + R}$	–	Mean of NDVI in a unit area
	Proportion of woodland and grass (PWG)	$PWG = \frac{N}{A}$	%	The proportion of forest land and grassland area in unit area
Artificial factors indicators	Population (POP)	$POP = P_A$	–	Number of population in unit area
	Building density (BD)	$BD = \frac{S_b}{A}$	%	The proportion of total building footprint to unit area
	Mean building height (MBH)	$MBH = \frac{\sum_{i=1}^{n} H_i}{n}$	m	Mean height of buildings in unit area
	Normalized difference built-up index (NDBI)	$NDBI = \frac{MIR - NIR}{MIR + NIR}$	–	Mean of NDBI in unit area
	Sky view factor (SVF)	$SVF = 1 - \sum_{i=1}^{n} sin^2\beta_i \frac{\alpha_i}{360^\circ}$	–	Mean of the visibility of the street sky in unit area
	Proportion of construction land (PCL)	$PCL = \frac{B}{A}$	%	The proportion of construction land area in unit area
	Proportion of soil bareness (PSB)	$PSB = \frac{S}{A}$	%	The proportion of soil bareness area in unit area
	Floor area ratio (FAR)	$FAR = \frac{\sum_{i=1}^{n}(F \times S)}{A}$	–	The proportion of total above ground floor area of building to unit area

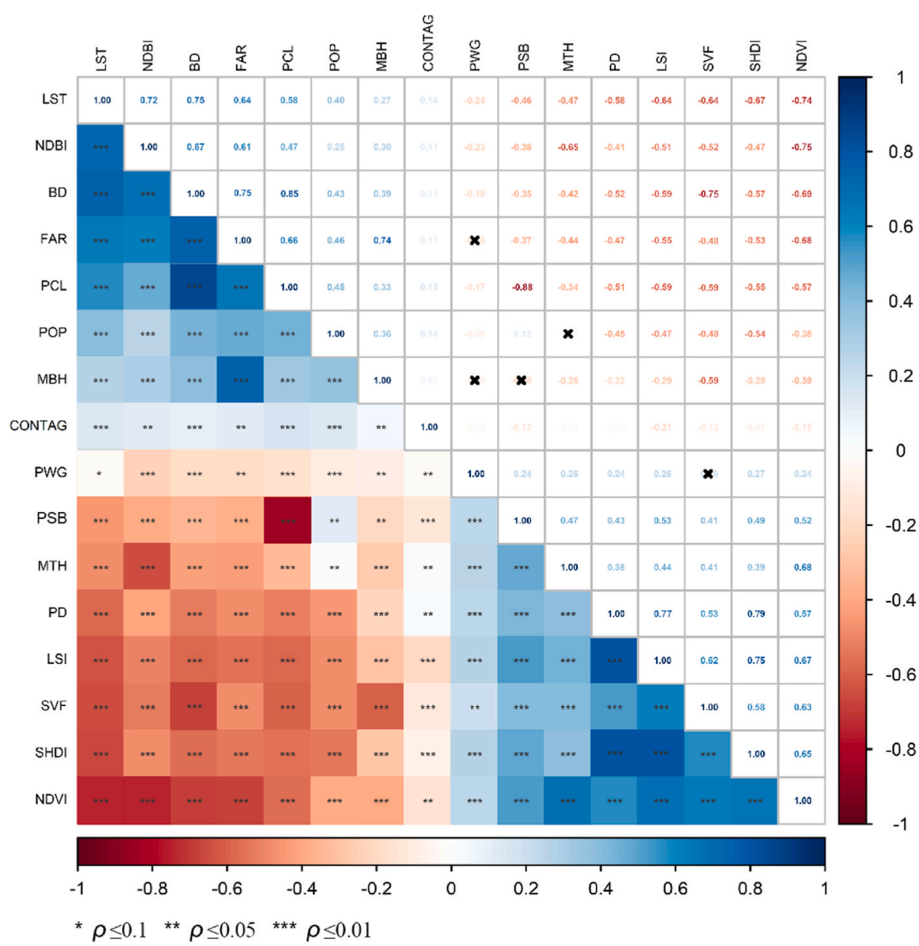

Fig. 4. The Spearman's correlation coefficient between UFIs and the LST.

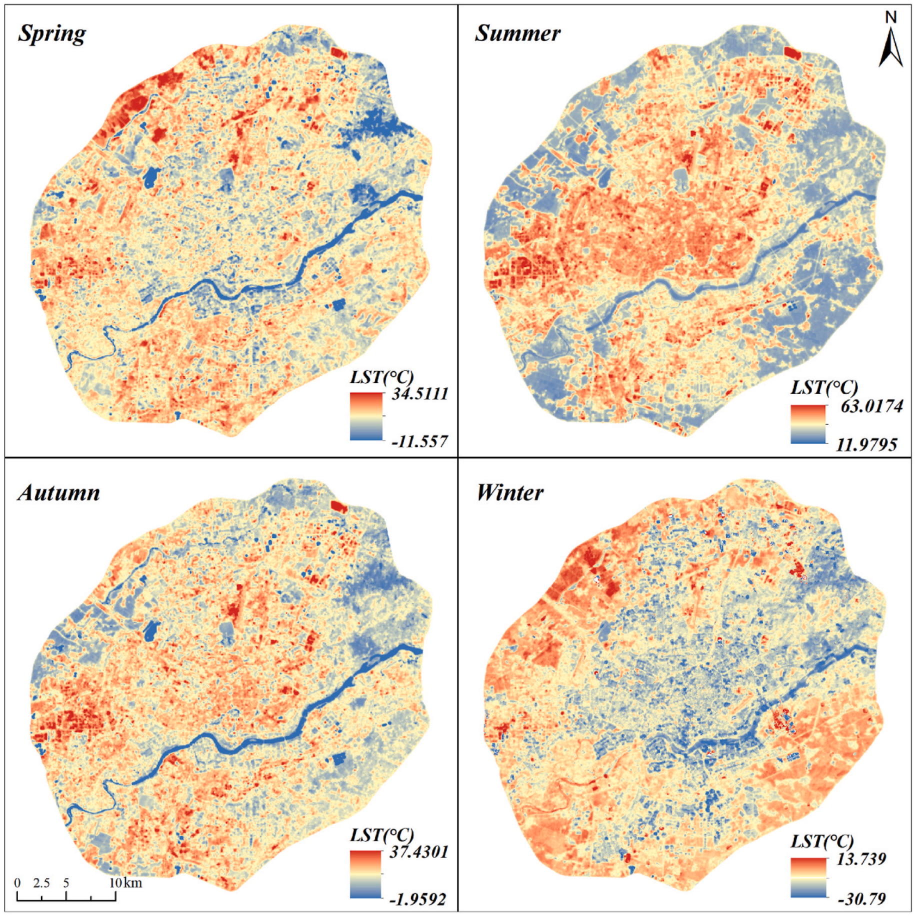

Fig. 5. Spatial distribution of seasonal LST in Shenyang.

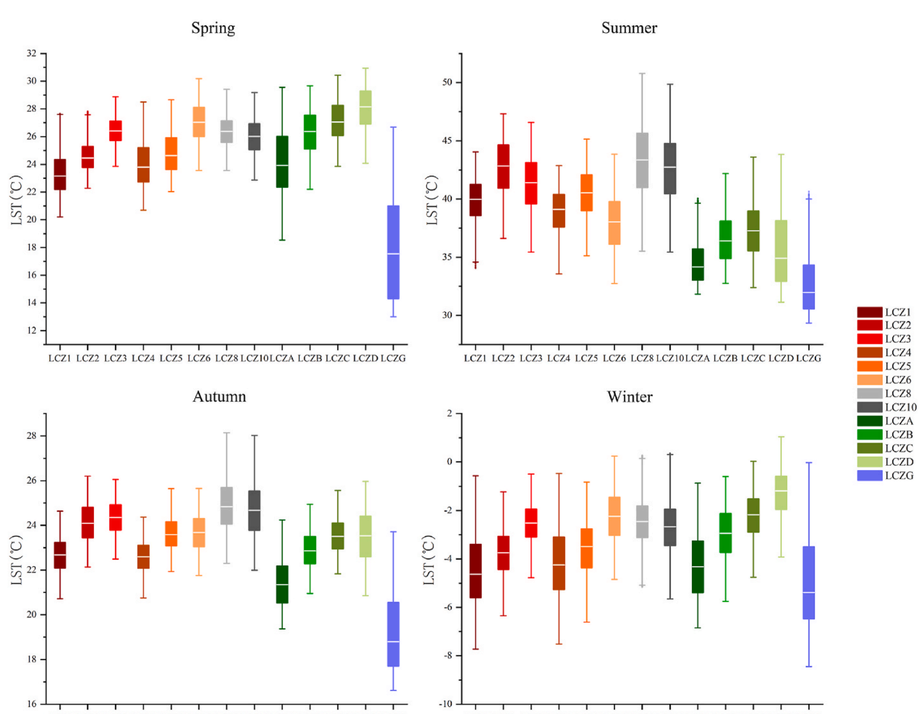

Fig. 6. Box plots depicting the seasonal LST in different LCZs.

individual LCZs, they represent only a small proportion of the total data. As shown in Fig. 6, interclass comparisons among the different LCZs revealed greater LST divergence, whereas intraclass comparisons showed smaller LST divergences in spring and autumn, and larger divergences in summer and winter.

For the built-type LCZs, the large buildings in LCZ8 absorb solar radiation more rapidly due to their substantial concrete surfaces, resulting in higher LST values. Additionally, ongoing industrial production contributes to the consistently high LST in LCZ10. In the land-cover-type LCZs, LCZG consistently exhibited the lowest mean LST values across all seasons, while LCZD displayed the highest mean LST values in all seasons except summer, which can be attributed to the seasonal vegetation cycle; maximum crop cover in summer, followed by harvest in autumn, dormant farmland in winter, and early germination in spring. Building height also affects the LST, with lower building heights and the associated shadows leading to mean LST values for the built-type LCZs falling in the order LCZ1 < LCZ2 < LCZ3 and LCZ4 < LCZ5 < LCZ6 from spring to winter. Similarly, as vegetation density and height decrease, the mean LST values for land-cover-type LCZs increase in the order LCZA < LCZB < LCZC < LCZD. In summer, building height remains constant but the form shifts from compact to open, and the mean LST values follow the pattern: LCZ1 > LCZ4, LCZ2 > LCZ5, and LCZ3 > LCZ6. The results show that the mean LST values are generally higher for built-type LCZs than land-cover-type LCZs across all seasons.

4.2. Differences in the effects of UFIs on the LST in different LCZs at grid scale

SHAP values were used to illustrate the relative importance of all UFIs on the LST across different LCZs, as shown in Fig. 7. As presented in Table 4, the XGBoost regression models achieved R^2 values ranging from 0.6 to 0.9 across most LCZ types, demonstrating strong explanatory power for the relationship between LST and UFIs. The relative significance of various UFIs differed notably among the different LCZs, and no single factor exhibited a dominant influence across all LCZ types simultaneously. However, where surface environments are similar in LCZ1–LCZ6 and LCZA–LCZD, NDBI and NDVI were found to be the primary factors influencing the LST. In LCZ1–LCZ6 (built-type LCZs), the contribution of MBH decreased progressively as the building height declined, while the contribution of BD increased. In LCZG, natural UFIs dominated overall contributions, accounting for at least 79.51 % in summer. During winter, natural factors dominated LST variation across LCZs, while artificial factors were more influential in other seasons. NDBI, NDVI, and SHDI were thus identified as the primary drivers of spatiotemporal heterogeneity in LST across all LCZ types.

4.3. Differences in the contributions of UFIs to the LST at two scales

The results of the XGBoost–SHAP model indicated that the impacts of the UFIs varied across the four seasons, following the trend: summer > spring > autumn > winter. From the block-scale perspective (Fig. 8), the total contributions of artificial and natural UFIs were as follows: spring (60.00 % and 40.00 %), summer (66.85 % and 33.15 %), autumn (43.84 % and 56.16 %), and winter (43.89 % and 56.11 %). Seasonally, the top-ranked indicator with the greatest impact in each season was associated with an increase in the LST; the larger the value, the higher the LST. The most impactful indicators across all seasons were the artificial factor indicators, with NDBI the most influential in spring and winter, and BD the most influential in summer and autumn. The top three indicators affecting the LST were NDBI > NDVI > MBH (spring), BD > NDVI > SHDI (summer), BD > NDBI > SHDI (autumn), and NDBI > NDVI > MBH (winter).

UFIs that showed positive correlations with LST included NDBI, BD, and FAR, whereas MBH, POP, MTH, and CONTAG exhibited negative correlations. NDVI had a cooling effect on LST in summer but a warming effect in the other seasons. PD exhibited a tri-phasic effect on LST—initial suppression, followed by promotion, and then suppression again, as its values progressively increased. LSI showed narrower SHAP value distributions (indicating limited influence) in summer and autumn but exerted warming effects in other seasons. SHDI produced pronounced cooling effects during summer and autumn but contributed minimally in spring and winter. BD induced significant warming in summer and autumn but had weaker effects in spring and winter. POP was negatively correlated with LST in spring and winter, whereas higher POP values increased LST in summer and autumn. CONTAG and MBH maintained consistent negative correlations with LST across all seasons. MTH showed a weak cooling effect on LST. SVF exhibited an inhibiting effect on LST at the block scale.

From the grid-scale perspective (Fig. 9), the total contributions of artificial and natural UFIs demonstrated the following trend: spring (66.03 % and 33.97 %), summer (76.65 % and 23.35 %), autumn (62.07 % and 37.93 %), and winter (47.11 % and 52.89 %). The top indicators with the greatest impact on LST were all artificial (NDBI) across all four seasons, with the top three metrics affecting the LST falling in the order

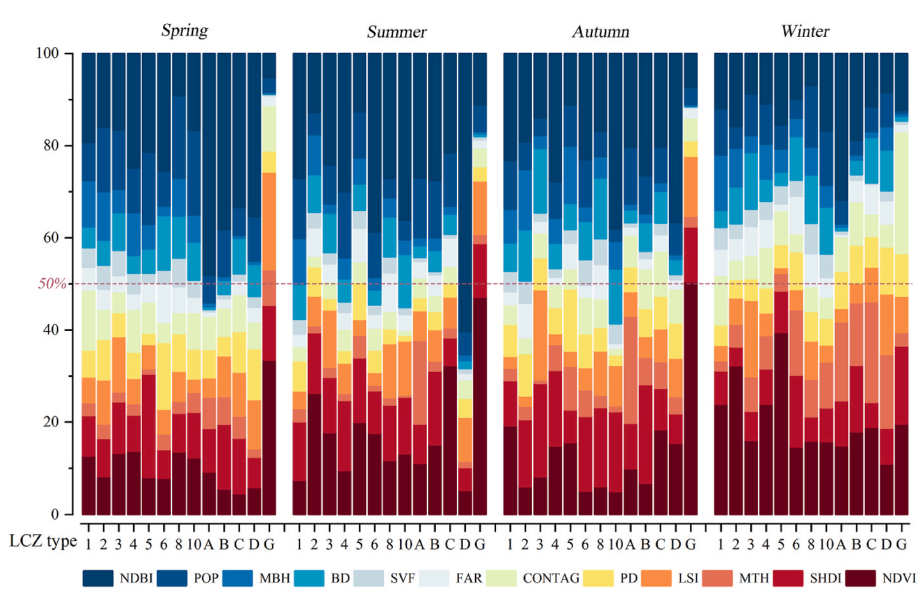

Fig. 7. Contributions of different UFIs to the seasonal LSTs in different LCZs.

Table 4
The R^2, MSE, and Time of XGBoost regression in different LCZs.

LCZ types	R^2				MSE				Time (s)			
Season	Spring	Summer	Autumn	Winter	Spring	Summer	Autumn	Winter	Spring	Summer	Autumn	Winter
LCZ1	0.804	0.742	0.697	0.763	0.023	0.018	0.030	0.025	0.650	0.120	0.100	0.440
LCZ2	0.736	0.859	0.738	0.653	0.026	0.018	0.034	0.025	0.930	0.950	0.960	1.630
LCZ3	0.505	0.819	0.643	0.505	0.034	0.019	0.036	0.030	0.810	0.100	0.290	0.930
LCZ4	0.841	0.800	0.645	0.743	0.022	0.019	0.035	0.026	0.430	0.120	0.290	0.440
LCZ5	0.819	0.828	0.561	0.695	0.022	0.019	0.039	0.025	0.140	0.110	0.090	0.240
LCZ6	0.659	0.884	0.623	0.699	0.030	0.016	0.038	0.027	1.010	0.120	0.970	0.110
LCZ8	0.665	0.862	0.740	0.578	0.025	0.015	0.027	0.024	1.200	0.870	0.970	1.200
LCZ10	0.620	0.799	0.725	0.589	0.030	0.018	0.028	0.025	1.140	0.290	0.310	0.900
LCZA	0.818	0.855	0.795	0.841	0.013	0.018	0.020	0.020	2.500	0.410	0.230	0.250
LCZB	0.805	0.827	0.699	0.705	0.020	0.020	0.033	0.027	0.530	0.100	0.110	0.550
LCZC	0.706	0.801	0.631	0.619	0.031	0.014	0.035	0.026	0.510	0.550	0.690	0.290
LCZD	0.757	0.860	0.869	0.724	0.029	0.009	0.024	0.024	0.880	0.710	0.710	0.950
LCZG	0.873	0.865	0.864	0.823	0.014	0.008	0.009	0.027	0.420	1.390	1.270	1.450

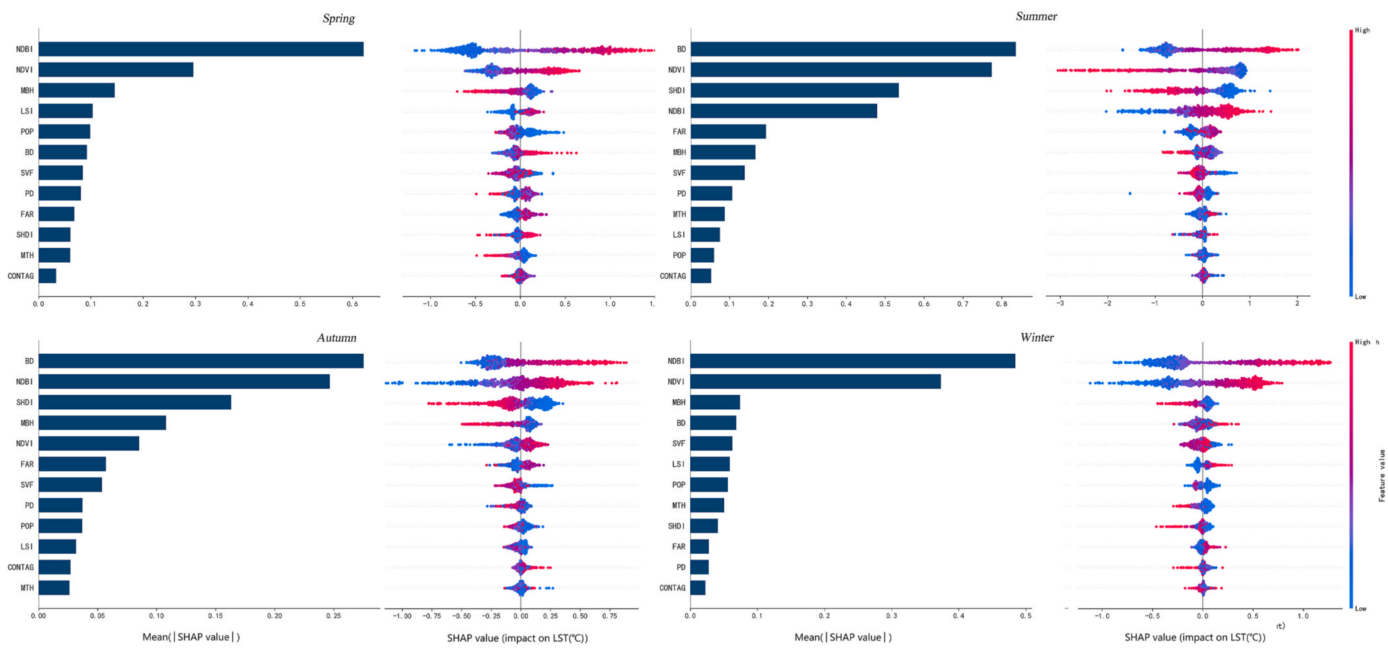

Fig. 8. Results of the XGBoost-SHAP model, depicting the effect of different UFIs on LST at the block scale.
The left panel shows the global importance of UFIs to LST changes, sorted from top to bottom by contribution degree. The right panel presents the local interpretation of UFIs' impacts on LST changes. A positive SHAP value indicates a warming effect, while a negative SHAP value represents a cooling effect. The x-axis denotes the range of cooling or warming (°C). The values of the features (UFIs) are represented by colors, with their range from low to high indicated by the color gradient bar on the right, which transitions from blue to red.

NDBI > POP > NDVI (spring), NDBI > NDVI > POP (summer), NDBI > SHDI > POP (autumn), and NDBI > NDVI > BD (winter).

CONTAG exhibited a consistent negative correlation with LST across all four seasons. NDVI demonstrated an inhibitory effect on LST in summer but showed the opposite pattern in the other seasons. The contribution of LSI to LST was significantly greater in summer than in other seasons, manifesting as a suppressive effect. PD showed a negative correlation with LST in spring and winter, while in the other seasons its influence was limited, as indicated by narrower SHAP value distributions and reduced impact magnitude. SHDI exhibited pronounced inhibitory effects on LST. BD contributed to distinct warming effects in summer and autumn but displayed variable behavior in spring and winter. POP correlated negatively with LST in spring and winter, whereas high values enhanced LST in summer and autumn. MBH demonstrated increasingly strong cooling effects with higher values in winter; in contrast, its influence in other seasons was more variable. SVF had a limited effect, as shown by the narrow distribution of SHAP values across all four seasons, and exhibited a facilitating effect on LST at the grid scale.

At both scales, NDBI exhibited a positive effect on LST, while NDVI showed a negative correlation with LST in summer and a positive correlation in all other seasons. The consistently high rankings and influence of NDBI, BD, POP, and MBH across all seasons indicate that human construction activities significantly impact the thermal environment of the city. Meanwhile, the persistently high rankings and influence of SHDI and NDVI suggest that natural vegetation remains a key regulator of temperature. It is worth noting that, although the influence of both FAR and SVF is generally lower at the grid scale as compared to the block scale, SVF exhibits a particularly striking scale reversal, inhibiting LST on the block scale but facilitating it on the grid scale.

4.4. Differences in the responses of LST to UFIs at two scales

According to the results of the SHAP summary plots for the impacts of UFIs on LST at the block and grid scales, five indicators that consistently ranked among the top contributors across all four seasons were

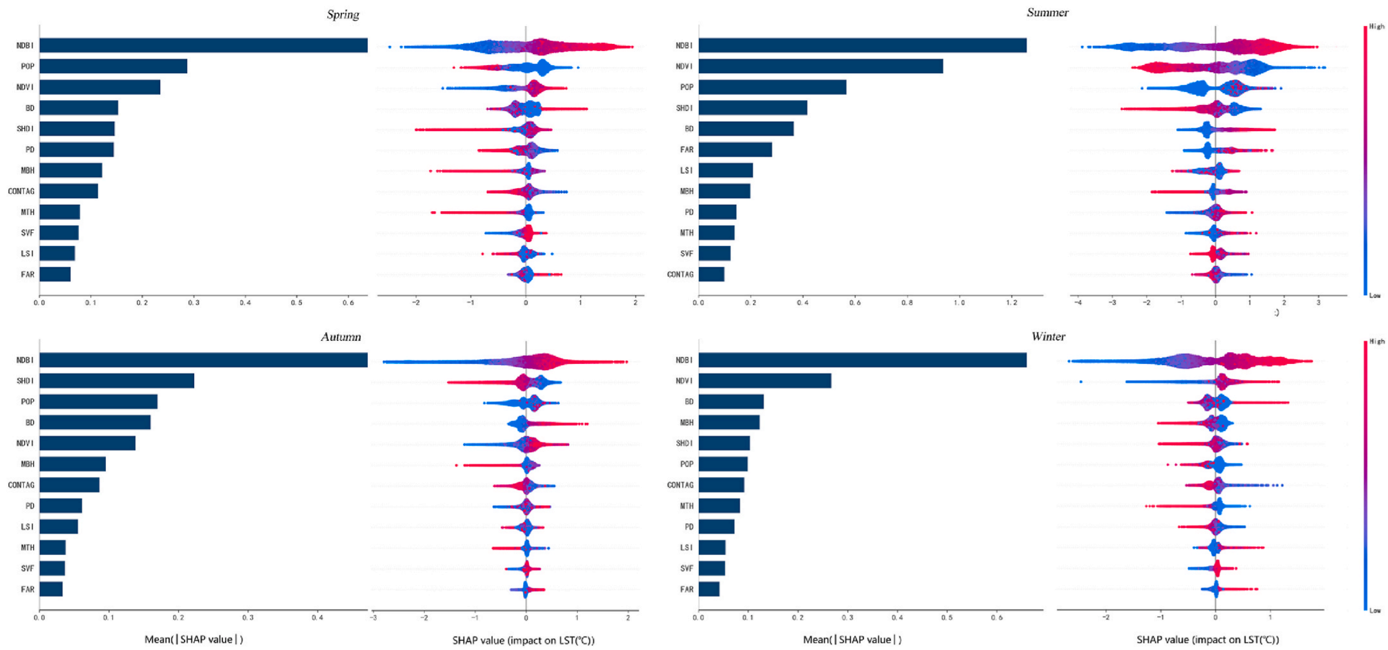

Fig. 9. Results of the XGBoost-SHAP model, depicting the effect of different UFIs on LST at the grid scale.

selected and an in-depth investigation of their influence on LST conducted on both scales (Figs. 10 and 11). Their total contribution to LST variation exceeded 60 % on both the block and grid scales in spring (67.05 % and 69.51 %), summer (79.34 % and 81.76 %), autumn (64.00 % and 73.42 %), and winter (64.07 % and 61.03 %).

The overall response trends of LST to UFIs were similar on both scales, featuring approximate and specific warming or cooling thresholds. As the NDBI increases, the SHAP value also increases. The NDBI threshold ranges from −0.05 to 0.05, representing the critical transition point at which urban landforms shift from natural to artificial landscapes, accelerating heat accumulation. A SHDI exceeding 0.65 produces a cooling effect on the LST.

However, the thresholds of some indicators vary across seasons. Specifically, the NDVI values above 0.4 in summer trigger a cooling effect on the surface temperature, while generally remaining below 0.4 in other seasons. Increases in the NDVI are associated with a rising LST, indicating that the cooling effect of NDVI may be constrained by other factors (e.g., the NDBI) during these periods. The relationship between BD and LST remains positively correlated in summer and autumn, with BD values above 20 % causing a warming effect. In contrast, BD exhibits a "U"-shaped relationship with LST in spring and winter, where BD of >20 % promotes warming after an initial decline. At the block scale, When MBH exceeds 13 m, it demonstrates a significant cooling effect at the block scale, which is typically associated with plots that also have high FAR and high BD. When the MBH increases from 10 m to 20 m, the resulting effective shading leads to a 1°C decrease in the predicted value of LST. At the grid scale, POP shows a cooling effect in spring and winter but a warming effect in summer and autumn when values exceed 30. High-contribution UFIs often exhibit the strongest interaction effects with artificial factor indicators. More pronounced LST variations are captured at the grid scale. When the NDBI increases from 0 to 0.2, it exhibits a warming effect of nearly 3°C. Once the NDVI exceeds 0.4, there is a 2°C decrease in the predicted LST. After the SHDI surpasses 0.65, it shows a cooling effect: the maximum cooling reaches over 2°C in summer, and even the minimum cooling in winter is 1.2°C. For BD, as it increases from 20 % to 60 % in summer, the predicted LST rises by 1.5°C; in autumn, this increase is at least 0.6°C.

Focusing on summer, when the interactive effects are most prominent, we identified a significant interaction between the NDVI and the NDBI: In non-built-up areas (NDBI <0), NDVI is the key factor contributing to cooling, allowing the cooling effect of vegetation to be fully exerted (Fig. 11a). Vegetation can effectively consume energy through transpiration, and its canopy provides shading to reduce the absorption of solar radiation by the ground surface. In built-up areas (NDBI >0), the cooling effect of vegetation is weakened; however, regions with high NDVI can still mitigate the warming intensity caused by NDBI (Fig. 11b). Additionally, high BD and high FAR lead to a significant warming effect (Fig. 11d2). Nevertheless, some areas with low BD and high FAR also exhibit a cooling effect. The reason is that when BD is low, a higher FAR indicates greater building height, which provides effective shading and promotes air circulation on the side of buildings. When MBH is less than 10 m, a higher BD results in a more pronounced warming effect (Fig. 10e2). At this height, sufficiently deep urban canyons have not yet formed between buildings, and the heat absorbed by compact buildings exposed to sunlight is difficult to dissipate. When MBH is constant, areas with higher BD have poorer ventilation conditions and lower vegetation coverage; the shading effect brought by high-rise buildings is insufficient to reduce the heat absorption of impervious surfaces.

5. Discussion

5.1. Temporal trends and spatial heterogeneity of LST in different LCZs

In this study, all four seasons were analyzed to achieve a comprehensive understanding of the urban thermal environment throughout the year. The results showed an opposite effect in winter and spring as compared to summer and autumn. The central urban area is colder than the peripheral areas, not only in winter but also in spring, exhibiting seasonal lag. This phenomenon may be attributed to the temperate continental climate in Shenyang (Chen et al., 2024a). The cold air masses from the northwest have a relatively short transport distance and dominate in winter and spring, lowering the air temperature and resulting in climatic conditions that are distinct from those observed in summer and autumn (Chen et al., 2024b). Additionally, the wind speed is higher over the farmland landscapes in the urban periphery as compared to the city proper, bringing in dry, cold air, and this phenomenon is more pronounced in the urban periphery (Zhang and Wu, 2025). The relatively cold Russian Arctic sea surface temperature in

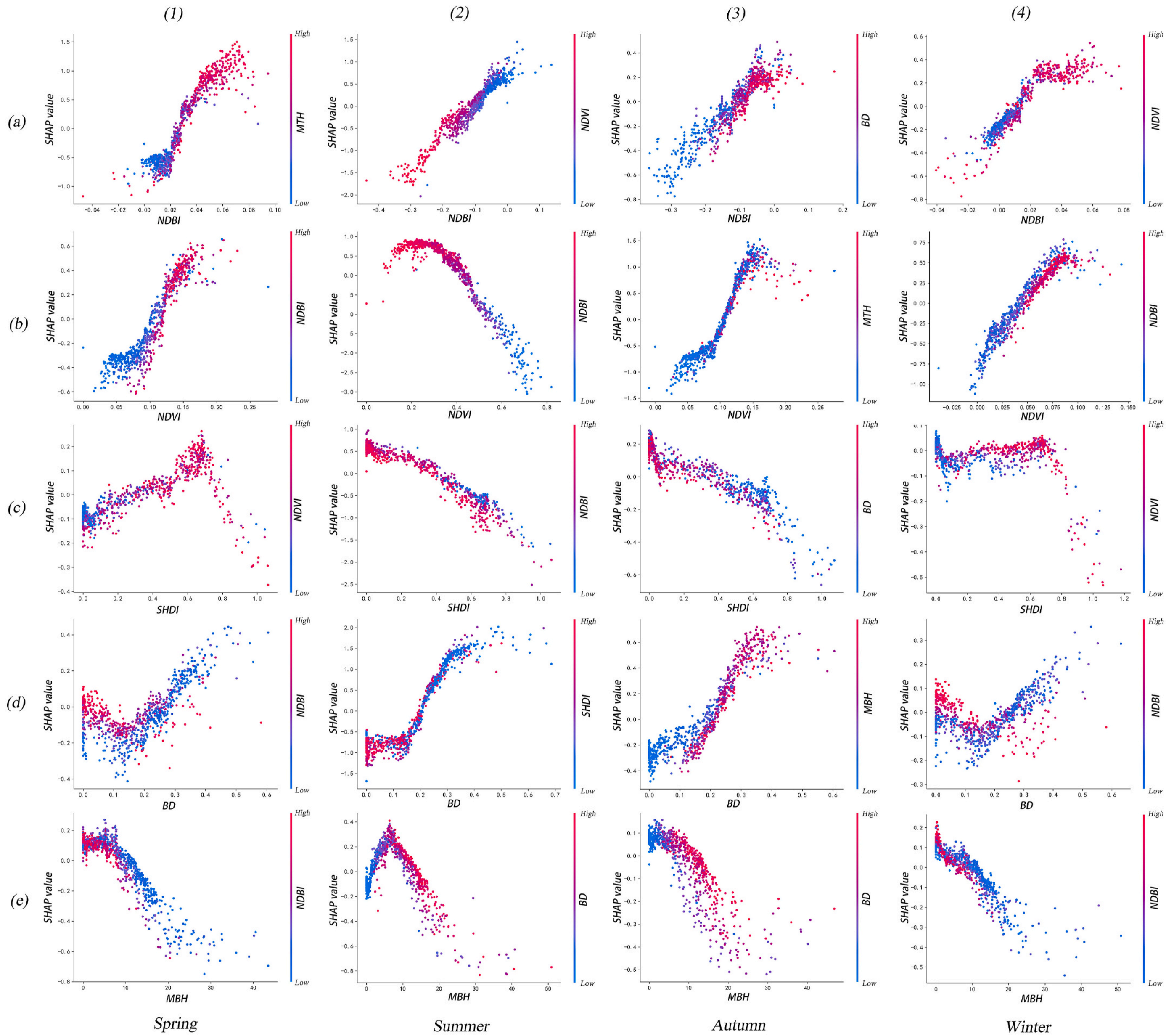

Fig. 10. Responses of LST to UFIs at the block scales.
LST is the dependent variable, and the x-axis represents the independent variables. Panels (a)–(e) denote LST's responses to the top six indicators with relatively high importance, while (1)– (4) represent spring, summer, autumn, and winter, respectively. The x-axis represents the independent variables, and the right side of the y-axis denotes the indicator that has the strongest interactive effect with the independent variable. Its values—ranging from low to high—are visualized using a color gradient from blue to red. This indicator interacts with the independent variable to exert a joint influence on the dependent variable (LST). Specifically, the value of the right side of the y-axis is expressed as the SHAP value (°C), which represents the predicted range of LST increase or decrease caused by changes in UFIs.

winter and spring weakens the role of the tropical Indo-Pacific dipole convection in warming the air temperature in Shenyang (Zhu et al., 2024). In urban areas, the shadows formed by the MBH and the rapid heat loss from buildings in cold weather result in the lower LST in the urban areas as compared to the peripheral regions (Cao et al., 2022).

Interestingly, the downstream segment of the Hunhe River demonstrated higher LST values than the upstream segment. This disparity is due to artificial interception within the city; the upstream water is calmer and more prone to freezing, resulting in LST values below 0 °C, while the water flowing in the downstream segment entrains a larger air concentration distribution and the relatively higher air temperature renders it less prone to freezing, leading to LSTs of above 0 °C (Wei et al., 2022). The interaction between water bodies and their surrounding riparian forest systems and the atmosphere reduces the LST in the adjacent areas (Jiao et al., 2024). While artificial water impoundment certainly secures sufficient water volume and surface area upstream, the downstream water bodies are more vulnerable to extreme weather. Under high-temperature conditions in summer, water evaporation increases, and the downstream area is highly prone to a reduced surface area, which, in turn, weakens the cooling effect on the thermal environment in the riparian regions (Khalaf and Rustum, 2023).

LCZ1–LCZ6 represent land use types dominated by built-up areas, while LCZA–LCZD represent land cover types dominated by vegetation cover. Where surface environments were similar, NDBI and NDVI were the primary factors influencing LST (Khan et al., 2024). The LST gradually decreased from LCZ1 to LCZ3 and from LCZ4 to LCZ6 in summer and winter, as the blocks became more open; however, the opposite trend was observed during spring and autumn, with higher LSTs in LCZ2

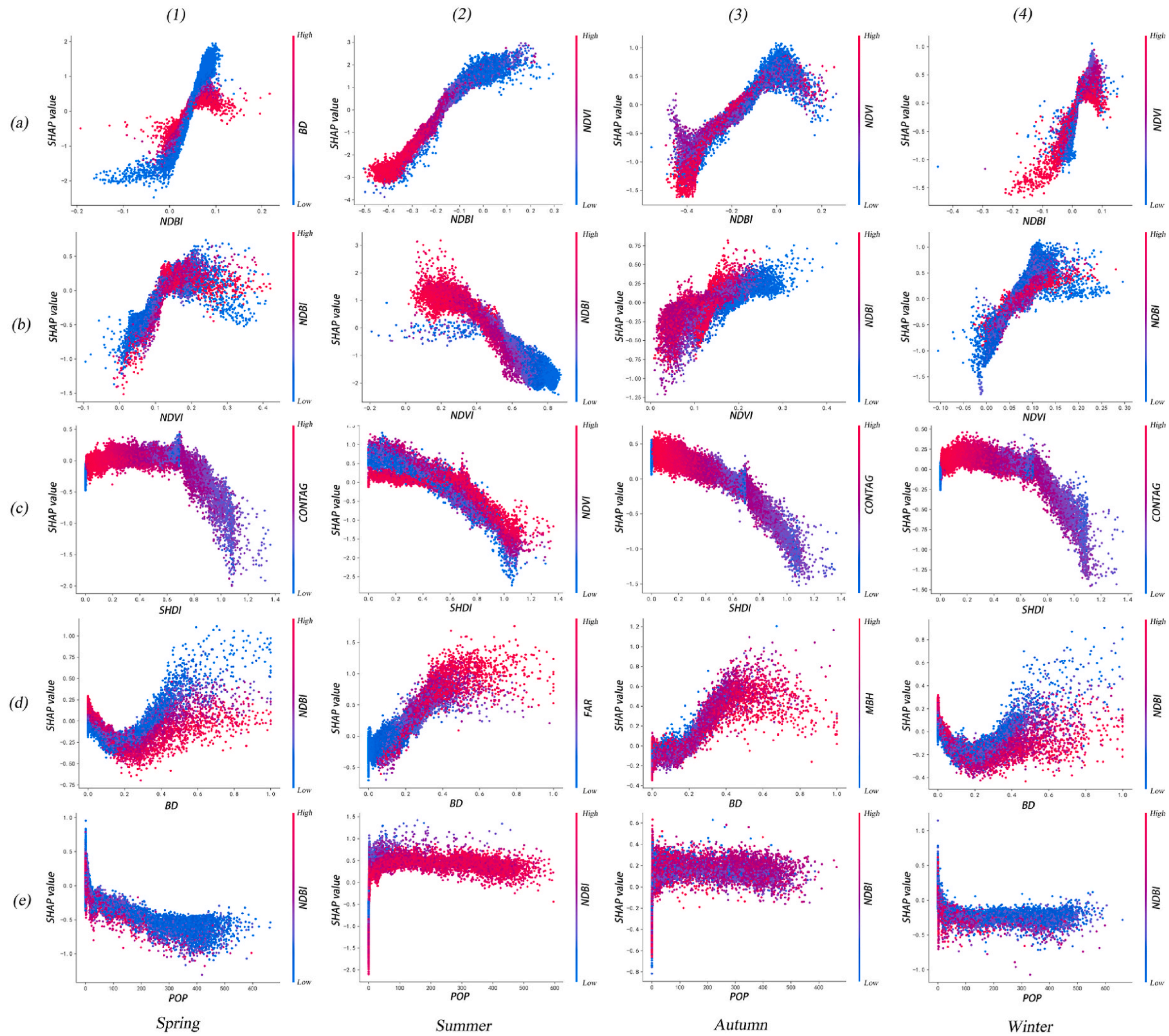

Fig. 11. Responses of LST to UFIs at the grid scales.

and LCZ5. The higher the vegetation coverage, the better the cooling effect (Pandey et al., 2024). Shenyang experiences a temperate continental climate with long, cold periods, and the observed suppression of the LST due to the presence of vegetation is weakened when it withers in the open blocks during spring and autumn. In summer, open neighborhoods with lush vegetation are less prone to heat accumulation, while in winter, open neighborhoods with dying trees and heavy snowpacks experience lower LST values, resulting in significant seasonal differences in terms of the LST (Xiang et al., 2022).

5.2. Analysis of the major factors that affected the LST

Cities exhibit complex morphological characteristics, characterized by an intricate fusion of artificial and natural elements. We therefore quantified the impact of various urban forms on the LST across different UFIs, with results indicating significant roles for NDVI and SHDI in reducing the LST. Statistical results indicated that the cooling effects of these factors are particularly pronounced in summer, consistent with the findings of You et al. (2021) and Naga Rajesh et al. (2023). Surface vegetation prevents the ground from direct exposure to solar radiation, reducing the absorption of heat by the surface (Yang et al., 2025), and trees planted around urban buildings cool the buildings by altering the air flow and increasing the rate at which the building energy is dissipated (Hui et al., 2024).

NDVI showed a negative correlation with LST in summer and a positive correlation in other seasons on both the block scale and grid scale. NDVI may not consistently suppress the LST due to seasonal fluctuation in its geophysical thermal efficiency (Xi et al., 2024). This is because the study area mainly comprises the central urban district, which is characterized by high building density, with most of the heat originating from the buildings and impervious surfaces. In addition, the cold season is long in Northeast China, and the average temperature in spring and autumn is relatively low. The higher the vegetation index value, the lower the LST. NDVI values are generally low except in summer; meanwhile, areas with a low NDVI are largely bare land, which, together with the buildings, significantly increases the LST (Pandey et al., 2024), and the limited green space cannot significantly alleviate the surface heat accumulation. A NDVI exceeding 0.4 will produce a notable cooling effect (Fig. 11b2). The vegetation is most vigorous in summer, when the vegetation coverage rate is much higher than that in other seasons, and the NDVI produces an obvious cooling effect when it exceeds a certain threshold. This study underscores the significance of natural UFIs in optimizing the urban thermal environment.

Regarding artificial UFIs, NDBI was found to be the primary factor influencing the LST in the region, with significant regression results observed in the summer. This finding aligns with the results of Guo et al. (2020) and Zhao et al. (2021). Urban development has led to the agglomeration of buildings, which increases the thermal conductivity and heat capacity of surface materials. Meanwhile, urban infrastructure construction uses a large number of impervious surface materials such as reinforced concrete and asphalt, altering Shenyang's original land cover and increasing the NDBI. These altered land parcels have a high capacity for heat absorption and storage, resulting in an increased LST (Gao et al., 2024). MBH was found to inhibit the LST; as the building height increased, the area of the shadow cast by the building on the surface also increased. The variation in the building height also affected the airflow between the buildings, leading to decreased LST (Lai et al., 2021), scale affecting its contribution. This is because as the scale expands, the number of buildings within a unit increases, and differences in the building height lead to the over-averaging of MBH values relative to the grid scale, resulting in the contribution of MBH being weaker on the block scale as compared to the grid scale (Chen et al., 2023). Overall, the LST was affected more by artificial than by natural elements, consistent with the results for 3D and 2D indices reported in previous studies (Alavipanah et al., 2018), which indicated that most 3D indices are linked to human activities. These findings collectively confirm the substantial influence of human construction on the urban thermal environment.

The research results are sensitive to changes in the scale of the study unit. SVF was found to inhibit LST at the block scale but facilitate it at the grid scale. This phenomenon may be attributed to a difference in the study scale, a certain buffer zone (smaller than a street canyon) is required for the SVF to obtain a scientifically meaningful descriptive indicator (Yuan et al., 2024). On the 100 m × 100 m grid scale, high SVF values tend to be distributed over bare land, open roads, and building rooftops because the underlying surfaces in these areas have strong heat absorption capacity, leading to a positive correlation between SVF and LST (Kim et al., 2022). However, at the block scale, high SVF values are usually found in park green spaces and small squares, where the underlying surfaces are dominated by vegetation and permeable bricks. These surfaces have a stronger cooling effect compared to densely built-up areas (with low SVF values) where concrete is the main material; thus, SVF and LST show a negative correlation (Deng et al., 2021). Comparison between the two scales has suggeste that the block scale is more effective in capturing urban form characteristics (Yin et al., 2018).

5.3. Advantages and improvements of XGBoost

Model selection prioritized two objectives: minimizing prediction errors to ensure analytical accuracy and optimizing computational efficiency for large-scale datasets. Performance was evaluated using three metrics: R^2 (Values closer to 1 indicate a better fit), MSE (lower values reflect reduced errors), and runtime (shorter durations reflect higher efficiency). We compared XGBoost, RF, and GBT under identical hardware conditions (Intel® Core™ i7-11700 @2.50 GHz, 32 GB RAM, NVIDIA RTX 3070 8 GB GPU), with all models tuned to optimal hyperparameters. These models were tested on eight datasets: four grid-scale seasonal sets (each >150,000 records) and four block-scale seasonal subsets (each >4200 records). Comparative analysis of R^2, MSE, and runtime revealed XGBoost's superior performance across all metrics. The complete experimental results are presented in Table 5, which confirmed the advantages of selecting XGBoost as the preferred model (Wan et al., 2025).

Notably, spatial autocorrelation and spatial dependence are crucial to the urban LST process. The spatial distribution of LST exhibits heterogeneity and is influenced by the natural and artificial characteristics of neighboring areas (Xu et al., 2024). To examine this, we conducted a spatial autocorrelation analysis on the residuals between the predicted and actual LST values. The results showed that Moran's I ranged from 0.24 to 0.3 ($p < 0.01$) at the block scale and from 0.31 to 0.39 ($p < 0.01$) at the grid scale. These values indicate a moderate positive spatial correlation, suggesting that the surrounding LST may either interfere with the impact of UFIs on LST or that there are unrecognized influencing factors yet to be explored. However, our comparative analysis at the 100 m × 100 m grid scale and block scale confirmed that the primary influencing factors are similar across the two spatial scales (Figs. 10 and 11). This finding verifies the robustness of the fundamental relationships we identified: the cross-scale common impacts of UFIs on LST can still provide valuable references for mitigating urban thermal environments. It should be noted that addressing spatial dependence requires higher-resolution data. In future research, the spatial detailed dependence of LST will need to be further explored using models such as the MGWR.

5.4. Implications for urban planning and management

It is impractical to extensively alter the LCZ types in built-up areas; therefore, urban planning measures should be tailored to local conditions, and thermal environment optimization policies should be implemented in a classified manner. These policies primarily target two types of areas: areas to be renewed in the city center and areas to be developed in new urban districts. The city center areas to be renewed mainly include two categories: old residential communities (LCZ2) and urban villages (LCZ3). For old residential communities (LCZ2), the primary measures include reducing the area of impervious pavement, increasing green space coverage, and promoting vertical greening. During the spatial layout adjustment of urban villages and newly built plots in new districts, the focus should be on constructing open-type LCZs (LCZ4–LCZ6), which involves increasing the vegetation coverage, planting large trees, and controlling the BD to an appropriate level of less than 40 %. Urban planning should strictly protect the main ventilation corridors from damage caused by newly developed buildings. In addition, for industrial parks that are dominated by LCZ8 and LCZ10, it is necessary to make full use of the large-area rooftop spaces, promote rooftop greening or photovoltaic rooftops, and strike a balance between the economic benefits and ecological benefits. LCZA–LCZC can effectively reduce LST, so urban parks should be well protected and extensive impervious pavement in parks avoided to lower the NDBI; meanwhile, diverse vegetation types should be selected to improve vegetation connectivity. Additionally, river area (LCZG) should be preserved and pocket parks appropriately arranged to increase urban vegetation

Table 5
The R^2, RMSE, and Time of XGBoost, RF and GBT model.

Evaluation index	Model	Grid scale				Block scale			
		Spring	Summer	Autumn	Winter	Spring	Summer	Autumn	Winter
R^2	XGBoost	0.78	0.8642	0.763	0.8132	0.663	0.8166	0.6504	0.7083
	RF	0.7716	0.8505	0.7545	0.7858	0.656	0.7994	0.6307	0.697
	GBT	0.7734	0.8526	0.7547	0.7877	0.6486	0.8101	0.6406	0.6954
RMSE(°C)	XGBoost	0.122	0.095	0.135	0.132	0.134	0.134	0.153	0.143
	RF	0.129	0.107	0.168	0.135	0.138	0.164	0.157	0.144
	GBT	0.126	0.106	0.137	0.134	0.139	0.136	0.155	0.145
Time(s)	XGBoost	5.38	5.71	5.62	5.3	0.37	0.08	0.41	0.10
	RF	77.07	71.59	53.02	119.8	3.74	2.17	2.24	10.63
	GBT	86.17	87.71	83.61	83.56	0.96	8.74	9.81	0.99

coverage.

5.5. Limitations

The indicators used in this study were selected based on the accessibility and general applicability of the data. However, cities exhibit complex physical characteristics and social dynamics that influence the urban thermal environment. Investigating thermal disparities among different LCZ categories requires consideration of certain variables that are difficult to quantify, such as building materials and local ventilation, which warrant further investigation.

While the LCZ classification provides a standardized framework for conducting horizontal comparisons between cities globally, the applicability of the study's findings to specific geographic regions depends on regional differences. Future cross-city and cross-regional studies should consider climatic and environmental differences in greater detail to account for interregional variations.

Currently, due to limitations in data accuracy, it is difficult to obtain and conduct large-scale studies on climate factors such as precipitation, wind, and circulation at the fine-grained urban scale. However, climate factors are important elements that cannot be ignored—they hold significant explanatory power for the seasonal variations in land surface temperature. Therefore, climate factors should be fully incorporated into consideration in future research.

6. Conclusion

This study developed detailed LCZ maps for the Fourth Ring Road area in Shenyang City. LST data for the study area were derived from Landsat 8 imagery and inverted using the RTE method, allowing for the analysis of spatiotemporal heterogeneity in LST across different LCZs. Regression analysis was conducted from both block and grid (100m × 100m) scale perspectives using the XGBoost model. The SHAP model was applied to interpret the impacts of UFIs on LST and to assess the responses of LST to UFIs. Furthermore, differences in the influence of UFIs across LCZ types were explored. The main conclusions are summarized below.

(1) LSTs across the four seasons within the Fourth Ring Road in Shenyang exhibited spatiotemporal heterogeneity. Surface temperatures in the city center were higher than in the periphery during summer and autumn, while the opposite trend was observed in spring and winter. LCZ8 and LCZ10 consistently exhibited relatively high average LSTs throughout the year. In spring, autumn, and winter, the mean LSTs of built-type LCZs followed the pattern LCZ1 < LCZ2 < LCZ3 and LCZ4 < LCZ5 < LCZ6 as building heights decreased. For land-cover-type LCZs, the pattern was LCZA < LCZB < LCZC < LCZD, reflecting sparser and shorter vegetation. In summer, for LCZs with similar building heights, mean LSTs followed LCZ1 > LCZ4, LCZ2 > LCZ5, and LCZ3 > LCZ6, as layouts shifted from compact to open.

(2) At the 100m × 100m grid scale, the contribution of UFIs to LST varied by LCZ type, with no single factor dominating across all types. However, NDBI and NDVI primarily influenced LST in LCZ1–6 and LCZA–D, which have similar surface environments. Natural UFIs dominated LCZG (≥79.51 % in summer). NDBI, NDVI, and SHDI were the primary drivers of spatiotemporal heterogeneity in LST across all LCZs.

(3) The XGBoost-SHAP model results reveal seasonal and scale-dependent impacts of UFIs on LST. The influence of UFIs follows the trend: summer > spring > autumn > winter. At the block scale, key seasonal influence of LST include NDBI in spring and winter, and BD in summer and autumn. UFIs positively correlated with LST include NDBI, BD, and FAR, while MBH, POP, MTH, and CONTAG show negative correlations. At the 100m × 100m grid scale, artificial UFIs dominate across all seasons (66.03 %–76.65 %), with NDBI being the primary driver of LST in every season. CONTAG maintains a consistently negative correlation with LST. SVF exhibits a scale reversal: it inhibits LST at the block scale but facilitates it at the grid scale. NDVI suppresses LST in summer but increases it in other seasons. Overall, artificial factor indicators (NDBI, BD, POP, and MBH) and natural factor indicators (SHDI and NDVI) are key thermal regulators.

(4) We selected five indicators that consistently ranked among the top contributors across all four seasons. Their total contribution to LST variation exceeded 60 % at both the block and 100m × 100m grid scales. LST responses to UFIs exhibited similar trends across scales, with distinct warming or cooling thresholds. NDBI's SHAP value increases with its magnitude, with the range of −0.05 to 0.05 representing a critical transition zone from natural to artificial landscapes, where building agglomeration accelerates heat accumulation. SHDI values greater than 0.65 had a cooling effect. Seasonal threshold differences were observed: lower NDVI values provided limited cooling, and significant cooling occurred only when NDVI exceeded 0.4, though NDVI remained generally low in all seasons except summer. BD was positively correlated with LST in summer and autumn, with warming effects evident when BD exceeded 20 %. In spring and winter, BD exhibited a U-shaped relationship with LST, showing warming effects when exceeding 40 %. High-contribution UFIs generally showed the most significant interactive effects with artificial factor indicators.

In conclusion, this study examined the spatiotemporal heterogeneity of LST across various LCZs and analyzed the influence of different UFIs on LST, highlighting the main drivers of this heterogeneity at both the block and 100m × 100m grid scales. The findings provide valuable insights for urban planners to develop effective policies at the block scale and optimize urban LCZ configurations. This approach can enhance the urban thermal environment and promote the development of healthy and livable urban spaces.

CRediT authorship contribution statement

Hongfei Li: Writing – review & editing, Methodology, Conceptualization. **Jun Yang:** Writing – review & editing, Writing – original draft, Software, Formal analysis, Data curation. **Jiaxing Xin:** Writing – review & editing, Data curation. **Wenbo Yu:** Writing – review & editing, Data curation. **Jiayi Ren:** Writing – review & editing, Data curation. **Huisheng Yu:** Writing – review & editing, Data curation. **Xiangming Xiao:** Writing – review & editing. **Jianhong (Cecilia) Xia:** Writing – original draft, Visualization, Investigation.

Software and data availability

Our software is implemented using Python 3.9 and leverages the sklearn library to build XGBoost-SHAP models. The Python code are available on the GitHub platform: https://github.com/liye369500/xgb-shap.git.

The data used in the article can be downloaded free of charge from the corresponding websites, the links to the data are provided in the manuscript.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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